

類別資料分析 第5章

The Analysis of Categorical Data

Chapter 5 : Building and Applying Logistic Regression Models

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Building and Applying Logistic Regression Models

※SAS code

```
options nodate nonotes ps=60;
data crab;
  input width cases satell;
cards;
22.69 14 5
23.84 14 4
24.77 28 17
25.84 39 21
26.79 22 15
27.74 24 20
28.67 18 15
30.41 14 14
;
```

```
proc genmod;
  model satell/cases=width/dist=bin
  link=logit obstats
  residuals waldci lrci;
proc logistic;
  model satell/cases=width/influence;
  output out=predict
  p=pi_hat lower=LCL upper=UCL;
proc print data=predict;
run;
```

Building and Applying Logistic Regression Models

※SAS output

The GENMOD Procedure			
Model Information		Parameter Information	
Description	Value	Parameter	Effect
Data Set	WORK.CRAB	PRM1	INTERCEPT
Distribution	<u>BINOMIAL</u>	PRM2	WIDTH
Link Function	<u>LOGIT</u>		
Dependent Variable	SATELL		
Dependent Variable	CASES		
Observations Used	8		
Number Of Events	111		
Number Of Trials	173		

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	6	G^2 5.9696	0.9949
Scaled Deviance	6	5.9696	0.9949
Pearson Chi-Square	6	χ^2 5.0292	0.8382
Scaled Pearson X2	6	5.0292	0.8382
Log Likelihood	.	-98.8470	.

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※SAS output

Analysis Of Parameter Estimates					
Parameter	DF	Estimate	Std Err	ChiSquare	Pr>Chi
INTERCEPT	1	$\hat{\alpha}$ -11.5140	2.5489	20.4056	0.0001
WIDTH	1	0.4647	0.0985	22.2337	0.0001
SCALE	0	$\hat{\beta}$ 1.0000	0.0000		

NOTE: The scale parameter was held fixed.

Wald statistic

Normal Confidence Intervals For Parameters		
Two-Sided Confidence Coefficient: 0.9500		
Parameter	Confidence Limits	
PRM1	Lower	-16.5098
PRM1	Upper	-6.5183
PRM2	Lower	0.2715
PRM2	Upper	0.6578

- 95% C. I. for α is (-16.5, -6.52)
- 95% C. I. for β is (0.27, 0.66)

Building and Applying Logistic Regression Models

※SAS output

Likelihood Ratio Based Confidence Intervals For Parameters

Two-Sided Confidence Coefficient: 0.9500				
Parameter	Confidence Limits		Parameter Values	
			PRM1	PRM2
PRM1	Lower	-16.7920	-16.7920	0.6688
PRM1	Upper	-6.7499	-6.7499	0.2815
PRM2	Lower	0.2810	-6.7614	0.2810
PRM2	Upper	0.6693	-16.7806	0.6693

y_i	n_i	$\hat{\pi}_i$	Observation Statistics				95% C. I. for $\pi(x)$	
SATELL	CASES	Pred	Xbeta	Std	HessWgt	(Lower	Upper)	
5	14	0.2748	-0.9705	0.3521	2.7899	0.1597	0.4303	
4	14	0.3927	-0.4361	0.2592	3.3387	0.2801	0.5180	
17	28	0.4990	-0.003956	0.2002	7.0000	0.4022	0.5959	
21	39	0.6209	0.4932	0.1723	9.1802	0.5388	0.6966	
15	22	0.7180	0.9347	0.1977	4.4542	0.6335	0.7895	
20	24	0.7984	1.3761	0.2569	3.8634	0.7053	0.8676	
15	18	0.8592	1.8083	0.3307	2.1781	0.7614	0.9210	
14	14	0.9319	2.6168	0.4854	0.8880	0.8410	0.9726	

$$\hat{\pi}_i = \frac{e^{\hat{\alpha} + \hat{\beta}x_i}}{1 + e^{\hat{\alpha} + \hat{\beta}x_i}}$$

Building and Applying Logistic Regression Models

※SAS output

$y_i - n_i \hat{\pi}_i$	Observation Statistics					
Resraw	Reschi	Resdev	StResdev	StReschi	Reslik	
1.1530	0.6903	0.6721	0.8309	0.8535	0.8388	
-1.4974	-0.8195	-0.8369	-0.9503	-0.9305	-0.9459	
3.0277	1.1444	1.1487	1.3544	1.3493	1.3530	
-3.2140	-1.0608	-1.0487	-1.2295	-1.2437	-1.2334	
-0.7966	-0.3774	-0.3729	-0.4103	-0.4153	-0.4112	
0.8391	0.4269	0.4370	0.5063	0.4946	0.5034	
-0.4648	-0.3149	-0.3075	-0.3523	-0.3608	-0.3543	
0.9529	1.0112	1.4049	1.5798	1.1371	1.4981	

Pearson
residual

Adjusted
residual

Building and Applying Logistic Regression Models

※SAS output

The LOGISTIC Procedure				
Data Set: WORK.CRAB		Response Profile		
Response Variable (Events): SATELL		Ordered Value	Binary Outcome	Count
Response Variable (Trials): CASES				
Number of Observations: 8				
Link Function: <u>Logit</u>				
		1	EVENT	111
		2	NO EVENT	62
Model Fitting Information and Testing Global Null Hypothesis BETA=0				
	Intercept	Intercept		
		and		
Criterion	Only	Covariates	Chi-Square for Covariates	
AIC	227.759	201.694		
SC	230.912	208.001		
-2 LOG L	225.759	197.694	28.064 with 1 DF (p=0.0001)	
Score			25.683 with 1 DF (p=0.0001)	

$$\begin{cases} H_0 : \text{logit } \pi = \alpha \\ H_1 : \text{logit } \pi = \alpha + \beta x \end{cases}$$

Building and Applying Logistic Regression Models

※SAS output

Analysis of Maximum Likelihood Estimates					
Variable	DF	Parameter Estimate	Standard Error	Wald Chi-Square	Pr > Chi-Square
INTERCPT	1	-11.5140	2.5489	20.4056	0.0001
WIDTH	1	<u>0.4647</u>	<u>0.0985</u>	22.2337	0.0001
Standardized Estimate					
					0.529092

Analysis of Maximum Likelihood Estimates	
Variable	Odds Ratio
INTERCPT	
WIDTH	1.592

Association of Predicted Probabilities and Observed Responses	
Concordant = 66.3%	Somers' D = 0.454
Discordant = 20.9%	Gamma = 0.520
Tied = 12.8%	Tau-a = 0.210
(6882 pairs)	c = 0.727

$$e^{\hat{\beta}} = e^{0.4647} = 1.592$$

$$\hat{\pi}_i$$

$$\begin{aligned} \cdot \pi &= \pi(x) = P(Y = 1 | X = x) \\ \cdot \log \frac{\pi}{1-\pi} &= \alpha + \beta x \\ \cdot \hat{\pi}_i &= \frac{e^{\alpha + \beta x_i}}{1 + e^{\alpha + \beta x_i}} \end{aligned}$$

Building and Applying Logistic Regression Models

※SAS output

Regression Diagnostics			
Covariates		Pearson Residual	
Case Number	WIDTH	Value	(1 unit = e_i)
1	22.6900	0.6903	
2	23.8400	-0.8195	
3	24.7700	1.1444	
4	25.8400	-1.0608	
5	26.7900	-0.3774	
6	27.7400	0.4269	
7	28.6700	-0.3149	
8	30.4100	1.0112	

Deviance Residual		(1 unit = 0.18)	
Case Number	Value	-8 -4 0 2 4 6 8	
1	0.6721		*
2	-0.8369	*	
3	1.1487		*
4	-1.0487	*	
5	-0.3729		*
6	0.4370		*
7	-0.3075		*
8	1.4049		*

$$e_i = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{n_i \hat{\pi}_i (1 - \hat{\pi}_i)}}$$

Building and Applying Logistic Regression Models

※SAS output

Hat Matrix Diagonal		(1 unit = 0.02)	
Case Number	Value	0 2 4 6 8 12 16	
1	0.3458		*
2	0.2244		*
3	0.2807		*
4	0.2726		*
5	0.1741		*
6	0.2551		*
7	0.2382		*
8	0.2092		*

$$e_i = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{n_i \hat{\pi}_i (1 - \hat{\pi}_i)}}$$

Building and Applying Logistic Regression Models

※SAS output

Regression Diagnostics			
$\hat{\alpha} - \hat{\alpha}^{(i)}$		INTERCPT	Dfbeta
Case Number	Value	(1 unit = 0.07)	
		-8 -4 0 2 4 6 8	
1	0.5604		*
2	-0.3956	*	
3	0.4776		*
4	-0.0365		*
5	0.0821		*
6	-0.2011	*	
7	0.1647		*
8	-0.5315	*	

WIDTH		Dfbeta	
Case Number	Value	(1 unit = 0.07)	
		-8 -4 0 2 4 6 8	
1	-0.5411	*	
2	0.3739		*
3	-0.4295	*	
4	-0.0150		*
5	-0.0936	*	
6	0.2147		*
7	-0.1722	*	
8	0.5468		*

$\hat{\alpha}^{(i)}$: 去掉第 i 個 observation · 重新 fit model $\text{logit } \pi = \alpha + \beta x$
所得到的 α 的估計量

Building and Applying Logistic Regression Models

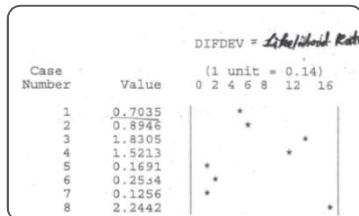
※SAS output

$ CI - CI^{(i)} $		C	
Case Number	Value	(1 unit = 0.04)	
		0 2 4 6 8 12 16	
1	0.3850		*
2	0.2504		*
3	0.7103		*
4	0.5796		*
5	0.0364	*	
6	0.0838	*	
7	0.0407	*	
8	0.3420		*

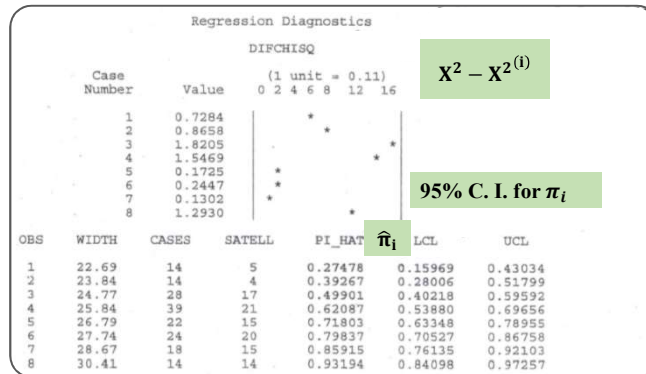
CBAR		C	
Case Number	Value	(1 unit = 0.03)	
		0 2 4 6 8 12 16	
1	0.2519		*
2	0.1943		*
3	0.5109		*
4	0.4216		*
5	0.0300	*	
6	0.0624	*	
7	0.0310	*	
8	0.2705		*

Building and Applying Logistic Regression Models

※SAS output



Likelihood Ratio difference



$$X^2 - X^2(i)$$

95% C. I. for π_i

OBS	WIDTH	CASES	SATELL	PI_HAT	$\hat{\pi}_i$	LCL	UCL
1	22.69	14	5	0.27478	0.15969	0.43034	
2	23.84	14	4	0.39267	0.28006	0.51799	
3	24.77	28	17	0.49901	0.40218	0.59592	
4	25.84	39	21	0.62087	0.53880	0.69656	
5	26.79	22	15	0.71803	0.63348	0.78955	
6	27.74	24	20	0.79837	0.70527	0.86758	
7	28.67	18	15	0.85915	0.76135	0.92103	
8	30.41	14	14	0.93194	0.84098	0.97257	

Building and Applying Logistic Regression Models

※SAS output

Table 5.8 Results of Fitting Several Logistic Regression Models to Horseshoe Crab Data

Model	Predictors	Deviance	DF	Models Compared	Difference	(Y, $\hat{\pi}$) Correlation
(1)	$C * S * W$	170.44	152	—	—	0.526
(2)	$C * S + C * W + S * W$	173.68	155	(2)-(1)	3.2 (df = 3)	
(3a)	$C * S + S * W$	177.34	158	(3a)-(2)	3.7 (df = 3)	
(3b)	$C * W + S * W$	181.56	161	(3b)-(2)	7.9 (df = 6)	
(3c)	$C * S + C * W$	173.69	157	(3c)-(2)	0.0 (df = 2)	
(4a)	$S + C * W$	181.64	163	(4a)-(3c)	8.0 (df = 6)	
(4b)	$W + C * S$	177.61	160	(4b)-(3c)	3.9 (df = 3)	
(5)	$C + S + W$	186.61	166	(5)-(4b)	9.0 (df = 6)	
(6a)	$C + S$	208.83	167	(6a)-(5)	22.2 (df = 1)	
(6b)	$S + W$	194.42	169	(6b)-(5)	7.8 (df = 3)	
(6c)	$C + W$	187.46	168	(6c)-(5)	0.8 (df = 2)	0.452
(7a)	C	212.06	169	(7a)-(6c)	24.5 (df = 1)	0.285
(7b)	W	194.45	171	(7b)-(6c)	7.0 (df = 3)	0.402
(8)	$C = \text{dark} + W$	187.96	170	(8)-(6c)	0.5 (df = 2)	0.447
(9)	None	225.76	172	(9)-(8)	37.8 (df = 2)	0.000

Note: C = color, S = spine condition, W = width

Building and Applying Logistic Regression Models

※某研究比較 338 位老兵使用 AZT 藥物延緩愛滋病免疫系統衰退的效果，得到下表

$$Y = \begin{cases} 1 & \text{衰退延緩} \\ 0 & \text{otherwise} \end{cases}$$

$$X = \begin{cases} 1 & \text{使用AZT} \\ -1 & \text{otherwise} \end{cases}$$

$$Z = \begin{cases} 1 & \text{white} \\ -1 & \text{black} \end{cases}$$

Z = 1		Y	
		1	0
X	1	14	93
	-1	32	81

Z = -1		Y	
		1	0
X	1	11	52
	-1	12	43

Building and Applying Logistic Regression Models

	Model 0	Model 1	Model 2	Model 3	Model 4
$\hat{\beta}_1 \pm sd(\hat{\beta}_1)$		-0.36 ± 0.14			
$\hat{\beta}_2 \pm sd(\hat{\beta}_2)$			0.044 ± 0.14		
G^2	0	6.93	0.1	6.97	8.35
df	0	1	1	2	3
D^2	8.35	1.42	8.25	1.38	0
df	3	2	2	1	0

在上述模式中，以 D^2 來看，model 1 及 model 3 不顯著

$$\therefore \begin{cases} H_0 : \text{logit } \pi = \alpha + \beta_1 x \\ H_1 : \text{logit } \pi = \alpha + \beta_1 x + \beta_2 z \end{cases} \quad \Delta D^2 = 1.42 - 1.38 = 0.04 < X^2_{1,0.05} \quad \therefore \text{Do not reject } H_0$$

故選擇 model 1 : $\text{logit } \pi = \alpha + \beta_1 x$

Building and Applying Logistic Regression Models

假設 $\pi_{ij} = P(Y = 1|X = i, Z = j)$

$$\cdot \text{Fixed } Z = 1 \quad \begin{cases} X = 1 & \text{logit } \pi_{1,1} = \alpha + \beta_1 + \beta_2 \\ X = -1 & \text{logit } \pi_{-1,1} = \alpha - \beta_1 + \beta_2 \end{cases} \quad \therefore \log \theta_{XY(1)} = e^{2\beta_1}$$

$$\cdot \text{Fixed } Z = -1 \quad \begin{cases} X = 1 & \text{logit } \pi_{1,-1} = \alpha + \beta_1 - \beta_2 \\ X = -1 & \text{logit } \pi_{-1,-1} = \alpha - \beta_1 - \beta_2 \end{cases} \quad \therefore \log \theta_{XY(2)} = e^{2\beta_1}$$

$\therefore$ 不管種族為何，有使用 AZT 對沒使用 AZT 的 odds ratio 為 $e^{-0.32} \approx 0.49$

Building and Applying Logistic Regression Models

結論：種族對衰退是否延緩並沒有影響，

無論種族為何，使用 AZT 的病人其衰退延緩的勝算是沒有使用 AZT 病人的一半（約 0.49 倍）

因此不使用 AZT 較好。

$$\hat{\pi}_{1,1} = 0.147 \quad \Rightarrow n_1 \cdot \hat{\pi}_{1,1} = 107 \cdot 0.147 = 15.729$$

$$\hat{\pi}_{-1,1} = 0.262 \quad n_2 \cdot \hat{\pi}_{-1,1} = 113 \cdot 0.262 = 29.606$$

$$\hat{\pi}_{1,-1} = 0.147 \quad n_3 \cdot \hat{\pi}_{1,-1} = 63 \cdot 0.147 = 9.261$$

$$\hat{\pi}_{-1,-1} = 0.262 \quad n_4 \cdot \hat{\pi}_{-1,-1} = 55 \cdot 0.262 = 14.41$$

Building and Applying Logistic Regression Models

∴ fitted table

		Y		
		1	0	
X	1	15.729	91.271	107
	-1	29.606	83.394	113

		Y		
		1	0	
X	1	9.261	53.739	63
	-1	14.41	40.59	55

$$\Rightarrow \theta_{XY(1)} = \frac{15.729 \times 83.394}{29.606 \times 91.271} = \frac{9.261 \times 40.59}{14.41 \times 53.739} = \theta_{XY(2)} \approx 0.49$$

Building and Applying Logistic Regression Models

※SAS code

```
options nodate nonotes ps=60;
data aids;
  input race $ azt $ yes no @@;
  cases=yes+no;
cards;
white y 14 93  white n 32 81
black y 11 52  black n 12 43
;
proc genmod order=data;
  class race azt;
  model yes/cases=race azt/dist=bin link=logit type3 obstats
    residuals;
run;
```

$$\text{Race } Z = \begin{cases} 1 & \text{white} \\ 0 & \text{black} \end{cases} \quad \text{AZT } X = \begin{cases} 1 & \text{use} \\ 0 & \text{no use} \end{cases}$$

$$\text{logit } \pi = \alpha + \beta_1 x + \beta_2 z$$

Building and Applying Logistic Regression Models

※SAS output

The GENMOD Procedure				
Model Information		Class Level Information		
Description	Value	Class	Levels	Values
Data Set	WORK.AIDS	RACE	2	white black
Distribution	<u>BINOMIAL</u>	AZT	2	y n
Link Function	<u>LOGIT</u>			
Dependent Variable	YES			
Dependent Variable	CASES			
Observations Used	4			
Number Of Events	69			
Number Of Trials	338			

Criteria For Assessing Goodness Of Fit				
Criterion	DF	Value	Value/DF	
<u>Deviance</u>	1	G² 1.3835	1.3835	1.3835
Scaled Deviance	1	1.3835	1.3835	1.3835
<u>Pearson Chi-Square</u>	1	X² 1.3910	1.3910	1.3910
Scaled Pearson X ²	1	1.3910	1.3910	1.3910
Log Likelihood	.	-167.5756	.	.

Building and Applying Logistic Regression Models

※SAS output

Analysis Of Parameter Estimates						
Parameter		DF	Estimate	Std Err	ChiSquare	Pr>Chi
INTERCEPT		1	-1.0736	0.2629	16.6705	0.0001
RACE	white	1	<u>0.0555</u>	0.2886	0.0370	0.8476
RACE	black	0	0.0000	0.0000	.	.
AZT	y	1	<u>-0.7195</u>	<u>0.2790</u>	<u>6.6507</u>	0.0099
AZT	n	0	0.0000	0.0000	.	.
SCALE		0	1.0000	0.0000	.	.

NOTE: The scale parameter was held fixed.

LR Statistics For Type 3 Analysis			
Source	DF	ChiSquare	Pr>Chi
RACE	1	0.0371	0.8473
AZT	1	6.8709	0.0088

Building and Applying Logistic Regression Models

※SAS output

$$\text{logit } \pi = \alpha + \beta_1 x + \beta_2 z$$

		y_i	n_i	$\hat{\pi}_i$			
		YES	CASES	Pred	Xbetā	Std	HessWgt
1	1	14	107	0.1496	-1.7375	0.2404	13.6144
1	0	32	113	0.2654	-1.0181	0.1985	22.0308
0	1	11	63	0.1427	-1.7930	0.2844	7.7073
0	0	12	55	0.2547	-1.0736	0.2629	10.4412

95% C. I. for $\pi(x)$		Observation Statistics			
(Lower	Upper)	Resraw	Reschi	Resdev	StResdev
0.0990	0.2199	-2.0098	-0.5447	-0.5547	-1.2010
0.1967	0.3477	2.0098	0.4282	0.4253	1.1713
0.0870	0.2252	2.0098	0.7239	0.7035	1.1461
0.1695	0.3640	-2.0098	-0.6220	-0.6326	-1.1995

$$y_i - n_i \hat{\pi}_i$$

Pearson
residual

Z	X	Y		
		1	0	
1	1	14	93	$n_1 = 107$
1	0	32	81	$n_2 = 113$
0	1	11	52	$n_3 = 63$
0	0	12	43	$n_4 = 55$

Observation Statistics	
StReschi	Reslik
-1.1794	-1.1841
1.1754	1.1784
1.1794	1.1670
-1.1794	-1.1850

Adjusted
residual

Building and Applying Logistic Regression Models

※SAS code

```
options nodate nonotes ps=60;
data aids;
  input race $ azt $ weaken $ wt;
cards;
white y y 14
white y n 93
white n y 32
white n n 81
black y y 11
black y n 52
black n y 12
black n n 43
;
proc catmod order=data;
  weight wt;
  population race azt;
  model weaken=;
run;
```

Building and Applying Logistic Regression Models

※SAS output

CATMOD PROCEDURE					
Response: WEAKEN			Response Levels (R)= 2		
Weight Variable: WT			Populations (S)= 4		
Data Set: AIDS			Total Frequency (N)= 338		
Frequency Missing: 0			Observations (Obs)= 8		
POPULATION PROFILES				RESPONSE PROFILES	
Sample	RACE	AZT	Sample Size	Response	WEAKEN
1	white	y	107	1	y
2	white	n	113	2	n
3	black	y	63		
4	black	n	55		

MAXIMUM-LIKELIHOOD ANALYSIS					
Iteration	Sub Iteration	-2 Log Likelihood	Convergence Criterion	Parameter Estimates	
0	0	468.56749	1.0000	0	
1	0	343.90163	0.2661	-1.1834	
2	0	342.12147	0.005176	-1.3522	
3	0	342.11761	0.0000113	-1.3606	
4	0	342.11761	6.873E-11	-1.3606	

Building and Applying Logistic Regression Models

※SAS output

MAXIMUM-LIKELIHOOD ANALYSIS					
Iteration	Sub Iteration	-2 Log Likelihood	Convergence Criterion	Parameter Estimates	
0	0	468.56749	1.0000	0	
1	0	343.90163	0.2661	-1.1834	
2	0	342.12147	0.005176	-1.3522	
3	0	342.11761	0.0000113	-1.3606	
4	0	342.11761	6.873E-11	-1.3606	

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE			
Source	DF	Chi-Square	Prob
INTERCEPT	1	101.66	0.0000
LIKELIHOOD RATIO	3	8.35	0.0393

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES					
Effect	Parameter	Estimate	Standard Error	Chi-Square	Prob
INTERCEPT	1	-1.3606	0.1349	101.66	0.0000

Building and Applying Logistic Regression Models

※SAS output

CATMOD PROCEDURE			
MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE			
Source	DF	Chi-Square	Prob
INTERCEPT	1	92.72	0.0000
RACE	1	0.09	0.7580
LIKELIHOOD RATIO	2	D^2 8.25	0.0161

$\begin{cases} H_0 : \text{logit } \pi = \alpha + \beta_2 z \\ H_1 : \text{Saturated model} \end{cases}$

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES					
Effect	Parameter	Estimate	Standard Error	Chi-Square	Prob
INTERCEPT	$\hat{\alpha}$ 1	1.3744	0.1427	92.72	0.0000
RACE	$\hat{\beta}_2$ 2	0.0440	0.1427	0.09	0.7580

Building and Applying Logistic Regression Models

※SAS output

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE			
Source	DF	Chi-Square	Prob
INTERCEPT	1	100.48	0.0000
AZT	1	6.71	0.0096
LIKELIHOOD RATIO	2	D^2 1.42	0.4915

$\begin{cases} H_0 : \text{logit } \pi = \alpha + \beta_1 x \\ H_1 : \text{Saturated model} \end{cases}$

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES					
Effect	Parameter	Estimate	Standard Error	Chi-Square	Prob
INTERCEPT	$\hat{\alpha}$ 1	-1.3970	0.1394	100.48	0.0000
AZT	$\hat{\beta}_1$ 2	-0.3609	0.1394	6.71	0.0096

Building and Applying Logistic Regression Models

※SAS output

ML PREDICTED VALUES FOR RESPONSE FUNCTIONS AND PROBABILITIES						
(x, x)	Sample Number	-----Observed-----		-----Predicted-----		Residual
		Function	Standard Error	Function	Standard Error	
1 1	1	-1.89354	0.286673	-1.75786	0.216556	-0.13568
	P1	0.130841	0.022501	0.147052	0.027163	-0.01622
	P2	0.869159	0.032601	0.852941	0.027163	0.016218
1 -1	2	-0.92871	0.208796	-1.03609	0.175476	0.107379
	P1	0.283186	0.042384	0.261905	0.033921	0.021281
	P2	0.716814	0.042384	0.738095	0.033921	-0.02128
-1 1	3	-1.55335	0.331873	-1.75786	0.216556	0.204509
	P1	0.174603	0.047829	0.147052	0.027163	0.027544
	P2	0.825397	0.047829	0.852941	0.027163	-0.02754
-1 -1	4	-1.27629	0.32648	-1.03609	0.175476	-0.2402
	P1	0.218182	0.05569	0.261905	0.033921	-0.04372
	P2	0.781818	0.05569	0.738095	0.033921	0.043723

Building and Applying Logistic Regression Models

※SAS output

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE				MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE			
Source	DF	Chi-Square	Prob	Source	DF	Chi-Square	Prob
INTERCEPT	1	91.82	0.0000	INTERCEPT	1	93.27	0.0000
RACE	1	0.04	0.8476	RACE	1	0.00	0.9899
AZT	1	6.65	0.0099	AZT	1	4.50	0.0338
				RACE*AZT	1	1.38	0.2399
LIKELIHOOD RATIO	1	1.38	0.2395	LIKELIHOOD RATIO	0	.	.

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES					
Effect	Parameter	Estimate	Standard Error	Chi-Square	Prob
INTERCEPT	1	-1.4056	0.1467	91.82	0.0000
RACE	2	0.0277	0.1443	0.04	0.8476
AZT	3	-0.3597	0.1395	6.65	0.0099

Building and Applying Logistic Regression Models

※SAS output

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES					
Effect	Parameter	Estimate	Standard Error	Chi-Square	Prob
INTERCEPT	1	-1.4130	0.1463	93.27	0.0000
RACE	2	0.00185	0.1463	0.00	0.9899
AZT	3	-0.3105	0.1463	4.50	0.0338
RACE*AZT	4	-0.1719	0.1463	1.38	0.2399

ML PREDICTED VALUES FOR RESPONSE FUNCTIONS AND PROBABILITIES						
Sample	Function Number	-----Observed-----		-----Predicted-----		Residual
		Function	Standard Error	Function	Standard Error	
1	1	-1.89354	0.286673	-1.89354	0.286673	0
	P1	0.130841	0.032601	0.130841	0.032601	0
	P2	0.869159	0.032601	0.869159	0.032601	0
2	1	-0.92871	0.208796	-0.92871	0.208796	0
	P1	0.283186	0.042384	0.283186	0.042384	0
	P2	0.716814	0.042384	0.716814	0.042384	0
3	1	-1.55335	0.331873	-1.55335	0.331873	0
	P1	0.174603	0.047829	0.174603	0.047829	0
	P2	0.825397	0.047829	0.825397	0.047829	0
4	1	-1.27629	0.32648	-1.27629	0.32648	0
	P1	0.218182	0.05569	0.218182	0.05569	0
	P2	0.781818	0.05569	0.781818	0.05569	0

observed value $y_i = n_i \hat{\pi}_i$ fitted value